

A Short Report on Simulation Based Inference

Scientific Machine Learning Seminar

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Abstract

Simulation Based Inference (SBI) offers a novel approach to solve inverse problems powered by machine learning. Hereby, a neural network learns to relate datapoints from a simulated dataset to a posterior distribution of model parameters. It can outperform classical Bayesian inference methods in inference speed and posterior accuracy. Additionally, SBI does not require any likelihood evaluation which often is computationally costly or has to be approximated. Common problems like model misspecification, neural inaccuracies or miscalibrated posteriors can limit the usefulness of SBI if they are not circumvented. Exemplary usecases for SBI are inferring the complex parameter posteriors from planetary microlensing events or gravitational waves from black hole mergers.

1 Introduction

What are inverse problems (in contrast to forward problems)

Inferring the parameter posterior with classical Bayesian methods or Simulation Based Inference

Short: examples where posterior estimation is needed

Section Roadmap

2 Classical Bayesian Inference

General concept of Bayes equation

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)}p(\theta) \tag{1}$$

Methods of inference, MCMC

Limitations: convergence, local approximation, multimodal posteriors

3 Introduction to Simulation Based Inference

Key concepts:

Skip Likelihood inference, implicit in simulation

Neural Posterior Estimation (NPE)

Sampling from simulator, train neural network, estimate posterior from observation Fig. 1

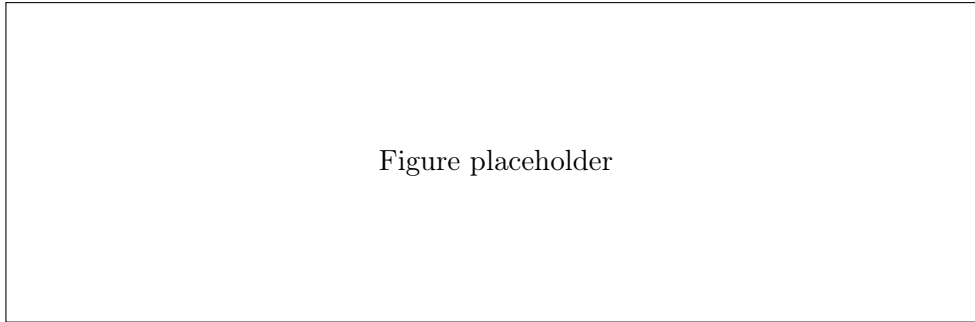


Figure 1: Workflow of SBI.

4 Simple SBI Example

Example: angled throw

choose parameter prior, create simulator of throw, simulate dataset
train neural network, here normalizing flows, any network works
loss function

$$L(\phi) = \tag{2}$$

use network to evaluate observation, data and param interval

5 Benefits and Limitations

Key benefits: very fast inference (Amortized Inference), Likelihood-free (no approximation or expensive computation), multimodal posteriors

Limitations: Upfront training, model misspecification (common pitfall) posterior is a neural estimation uncalibrated posterior

6 Application

Short explanation of microlensing

Results of usage of SBI ([Zhang et al., 2021](#))

Short explanation of gravitational waves and black hole mergers

use of flow matching networks

results of usage here ([Dax et al., 2023](#))

Other applications

7 Conclusion

Benefits and Limitations (stuff from abstract)

When to use SBI inverse problem, time critical inference, expensive likelihood evaluation, multimodal posterior

References

Maximilian Dax, Jonas Wildberger, Simon Buchholz, Stephen R. Green, Jakob H. Macke, and Bernhard Schölkopf. Flow matching for scalable simulation-based inference, 2023. URL <https://arxiv.org/abs/2305.17161>.

Keming Zhang, Joshua S. Bloom, B. Scott Gaudi, François Lanusse, Casey Lam, and Jessica R. Lu. Real-time likelihood-free inference of roman binary microlensing events with amortized neural posterior estimation. *The Astronomical Journal*, 161(6):262, May 2021. ISSN 1538-3881. doi: 10.3847/1538-3881/abf42e. URL <http://dx.doi.org/10.3847/1538-3881/abf42e>.